

Probability
Chapter 4 - Module 2

Random Walk Markov Chains

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June 15, 2026

In this module we study the basic random walk. We define a one-dimensional Random Walk Markov chain with transition probabilities

$$P_{i,i+1} = p \text{ and } P_{i,i-1} = 1 - p \text{ for } i = 0, \pm 1, \pm 2, \dots$$

with $0 < p < 1$.

We note first that every state communicates with every other state. In particular

$$P_{i,i+n}^n = p^n > 0 \text{ and } P_{i,i-n}^n = (1 - p)^n > 0 \text{ for } n = 1, 2, \dots$$

We conclude that the Markov chain is irreducible. Therefore every state is recurrent or every state is transient.

Theorem.

If $p = 1/2$, then every state in the Random Walk Markov Chain is recurrent.

If $p \neq 1/2$, then every state is transient.

The key ideas are:

A state i is recurrent if $\sum_{n=1}^{\infty} P_{ii}^n = +\infty$.

A state i is transient if $\sum_{n=1}^{\infty} P_{ii}^n < +\infty$.

Since the chain is irreducible, it is sufficient to show that

$$\sum_{n=1}^{\infty} P_{00}^n = +\infty \text{ or that } \sum_{n=1}^{\infty} P_{00}^n < +\infty.$$

We next note that the chain is periodic with period two. If you are in state i at index n , then you are in state $i + 1$ or $i - 1$ at index $n + 1$ with probability 1.

If you begin in state i , at index n , then you can only return to index i at indices $n + 2k$ for some k . So

$$P_{00}^{2n-1} = 0.$$

for any natural number n .

We now compute the exact probability that given that we start in state 0 that we return to state 0 in exactly $2n$ steps. This can only occur if we take n steps to the left and n steps to the right.

So let's define moving to the right as a "success" with probability p and moving to the left as a "failure" with probability $1 - p$. Then the number of successes in $2n$ trials is a binomial random variable X with

$$P(X = k) = \binom{2n}{k} p^k (1 - p)^{2n-k}.$$

We are interested in the case that $n = k$ in which case

$$P_{00}^{2n} = \binom{2n}{n} p^n (1 - p)^n \quad \text{for } n = 1, 2, \dots$$

since $2n - n = n$.

Recall that $n! \sim n^{n+1/2} e^{-n} \sqrt{2\pi}$ in the sense that

$$\lim_{n \rightarrow \infty} \frac{n^{n+1/2} e^{-n} \sqrt{2\pi}}{n!} = 1.$$

So

$$\begin{aligned} P_{00}^{2n} &= \binom{2n}{n} p^n (1-p)^n \\ &= \frac{(2n)!}{(n!)^2} p^n (1-p)^n \\ &\sim \frac{(2n)^{2n+1/2} e^{-2n} \sqrt{2\pi}}{(n^{n+1/2})^2 (e^{-n})^2 (\sqrt{2\pi})^2} (p(1-p))^n \\ &= \frac{2^{2n} \sqrt{2}}{\sqrt{2\pi}} \frac{n^{1/2}}{n} (p(1-p))^n \\ &= \frac{(4p(1-p))^n}{\sqrt{n\pi}}. \end{aligned}$$

Theorem. Let $\{a_n\}$ and $\{b_n\}$ be sequences of positive numbers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1.$$

Then

$$\sum_{n=1}^{\infty} a_n < \infty \text{ if and only if } \sum_{n=1}^{\infty} b_n < \infty.$$

For $p = 1/2$,

$$\frac{(4p(1-p))^n}{\sqrt{n\pi}} = \frac{1}{\sqrt{n\pi}}$$

and

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n\pi}} = \frac{1}{\sqrt{\pi}} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = +\infty.$$

since

$$\sum_{n=1}^{\infty} \frac{1}{n^s} < \infty$$

if and only if $s > 1$.

So when $p = 1/2$,

$$\begin{aligned}\sum_{n=1}^{\infty} P_{00}^{2n} &= \sum_{n=1}^{\infty} \binom{2n}{n} p^n (1-p)^n \\ &= \frac{1}{\sqrt{\pi}} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \\ &= \infty.\end{aligned}$$

So state 0 is recurrent. And since the Markov chain is irreducible, then every state is recurrent.

We now show that if $p \neq 1/2$, then every state is transient.

We note that the function $4p(1-p)$ is maximized when $p = 1/2$ and then $4p(1-p) = 1$.

For any other value of p , $0 < p < 1$, we have that $r = 4p(1-p) < 1$.

Therefore for $p \neq 1/2$,

$$\sum_{n=1}^{\infty} \frac{(4p(1-p))^n}{\sqrt{n\pi}} < \sum_{n=1}^{\infty} \frac{r^n}{\sqrt{n\pi}} < \sum_{n=1}^{\infty} r^n = \frac{1}{1-r} < \infty.$$

So

$$\sum_{n=1}^{\infty} P_{00}^{2n} < \infty$$

and so state 0 is transient and so every state is transient.

We extend the one dimensional random walk with $p = 1/2$ being the probability of moving left or right, to a two dimensional Markov chain with transition probabilities

$$P_{(i,j),(i,j+1)} = P_{(i,j),(i,j-1)} = P_{(i,j),(i+1,j)} = P_{(i,j),(i-1,j)} = 1/4.$$

So with this random walk, one either moves left or right or up or down all with equal probabilities.

We now show that every state is recurrent. And once again since every state communicates with every state, we can show that every state is recurrent by showing that $(0,0)$ is recurrent. To simplify the notation we write P_{00} for $P_{(0,0),(0,0)}$.

To get back to zero, one must take an even number of steps, so clearly $P_{00}^{2n-1} = 0$.

If the number of steps is say $2n$, then one can get back to zero by taking i steps to the left and to the right and by taking $n - i$ steps up and down. The probability of randomly having i steps left, i steps right, $n - i$ steps up and $n - i$ steps down is given by the multinomial distribution and the probability is

$$\frac{(2n)!}{i! i! (n - i)! (n - i)!} \left(\frac{1}{4}\right)^i \left(\frac{1}{4}\right)^i \left(\frac{1}{4}\right)^{n-i} \left(\frac{1}{4}\right)^{n-i}$$

$$= \frac{(2n)!}{i! i! (n - i)! (n - i)!} \left(\frac{1}{4}\right)^{2n}$$

for $i = 0, 1, \dots, n$.

So

$$P_{00}^{2n} = \sum_{i=0}^n \frac{(2n)!}{(i!)^2 ((n - i)!)^2} \left(\frac{1}{4}\right)^{2n}.$$

First of all we note that after a little combinatorial algebra that

$$\begin{aligned} P_{00}^{2n} &= \sum_{i=0}^n \frac{2n!}{(i!)^2 ((n-i)!)^2} \left(\frac{1}{4}\right)^{2n} \\ &= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \binom{2n}{n}. \end{aligned}$$

Once again we use the Stirling's approximation, which is $n! \sim n^{n+1/2} e^{-n} \sqrt{2\pi}$.

If we apply this to $(2n)!$ we have

$$(2n)! \sim (2n)^{2n+1/2} e^{-2n} \sqrt{2\pi}.$$

Then

$$\begin{aligned} \binom{2n}{n} &= \frac{(2n)!}{n!n!} \\ &\sim \frac{(2n)^{2n+1/2} e^{-2n} \sqrt{2\pi}}{(n^{n+1/2})^2 (e^{-n})^2 (\sqrt{2\pi})^2} \\ &= \frac{4^n}{\sqrt{n\pi}}. \end{aligned}$$

So

$$\begin{aligned}P_{00}^{2n} &= \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \binom{2n}{n} \\&\sim \left(\frac{1}{4}\right)^{2n} \frac{4^n}{\sqrt{n\pi}} \frac{4^n}{\sqrt{n\pi}} \\&= \frac{1}{n\pi}.\end{aligned}$$

Then clearly

$$\sum_{n=1}^{\infty} P_{00}^{2n} = \sum_{n=1}^{\infty} \frac{1}{n\pi} = +\infty.$$

Therefore state $(0,0)$ is recurrent and so every state is recurrent.

Suppose that not all of the transition probabilities are equal. In particular suppose that the probability that moving left is not equal to the probability of moving right. Ignoring the y component of the motion, we have from the previous analysis that the expected number of times that the x coordinate is zero is a finite number. So returning to the y axis is a transient phenomena. So $(0, 0)$ is also transient.

The analysis in the case that the probability of moving up and down are not equal also leads to the conclusion that every state is transient.

Question: Is every state recurrent if the probability of moving left equals the probability of moving right and the probability of moving up is the same as the probability of moving down in the case that not all of the probabilities are equal?

Finally note that the 3 and higher dimensional random walk with equal probabilities of moving in any direction are all equal (say equal to $1/8$ in every dimension in three dimensions) is an irreducible Markov chain in which every state is transient.